

Moving AI agents from sidebars onto visual canvases as collaborative entities unlocks new interaction paradigms where agents work spatially alongside users rather than replacing them.

Tldraw Evolution

- Tldraw is a London-based startup building an online whiteboard, SDK for canvas-based apps (used by Replit, Luba AI, Stitch), and a hackable React-based canvas runtime
- Make Real (2023) was an early breakthrough: draw UI mockups, send to vision model, get working HTML prototypes — first time non-technical users could create technical artifacts without coding
- Vision models struggle with spatial data because training conflicts (y-axis goes up in graphs but down in web coordinates, left vs stage-left ambiguity) make structured output harder than text generation

In 2023 Make Real was all the rage because there was no Lovable, vibe coding hadn't been coined — this was one of the first projects where non-technical people could make technical stuff without looking at code

From One-Shot to Agentic

- Early AI canvas integration was one-shot commands: 'draw a cat,' 'finish my slides,' 'complete this graph' — AI used structured outputs to create native canvas objects (shapes, text) rather than images
- Agentic loop introduced iterative refinement: AI produces output, reviews it, iterates until satisfied — borrowed conventions from coding agents (rejection, visible thinking, progress tracking)
- Still felt like handing keyboard to AI rather than true collaboration; worked well for education (tutoring, form completion, D&D character sheets) and design tools but lacked spatial presence

Fairies: Spatial Multi-Agent System

- Fairies are visible agents on canvas — draggable entities that show state, thinking, and actions spatially relative to work and each other (like multiple terminal windows but visual)
- Multiple fairies can work independently or collaboratively: one elected as leader to scout canvas, create to-do list, delegate tasks while observing and judging completion
- Solved agent orchestration challenges: shared state management, leader-follower patterns, preventing overlap when agents are blind during execution — figured out in parallel with broader industry (Oct-Dec 2024)
- Available at fairies.tldraw.com as accessible introduction to multi-agent systems: draw something, summon fairies, watch them collaborate on wireframes or complex tasks

Fairies is a great way to talk about multiple agents working together — they can do real work

Desktop App: Unrestricted Agency

- Electron wrapper with HTTP endpoint accepts JavaScript and executes it — terrible idea for web apps but safe for offline file-based desktop apps (worst case: user hurts themselves, not others)
- Enables bidirectional code-canvas sync: visualize code as diagrams, update diagrams to modify code, raise or lower abstraction level on demand
- AI writes code against editor API to add interactivity primitives that don't natively exist (hover, click handlers) — comfortable with script injection and modifying desktop apps (removed podcasts from Spotify by editing minified bundles)
- Local-first file-based apps with agent access shift from idealistic curiosity to practical necessity: maximizing agency requires handing sharp tools to users with full local control

I can almost tell that it would love to do this to websites — like just let me get my claws in there

Design Philosophy

- Agents should be collaborators working alongside users spatially, not replacements operating in sidebars — spatial presence makes state, progress, and coordination visible
- Safety boundaries determined by deployment context: sandboxed web apps need restrictions, local desktop apps can accept risk in exchange for capability
- File-over-app and local-first principles become practical requirements when maximizing agent capabilities — cloud-connected local execution is the architecture that enables unrestricted agency